Concurrency Anomalies



What are they?

Situations with undesirable concurrent behavior.

Want to prevent, but can allow to increase concurrency.

Dirty read / Read uncommitted



A's value depends on uncommitted value.

Due to WR conflict.

Allow dirty reads?

OK

- Concurrent writes unlikely
  - Writes = software updates
- Can redo if necessary
  - Airplane seat picker
  - Web searches
- Can accept a little inconsistent data
  - Statistical approximations

Not OK

- Dirty data used for updates.
- Data must be balanced (e.g., financial)

Unrepeatable read

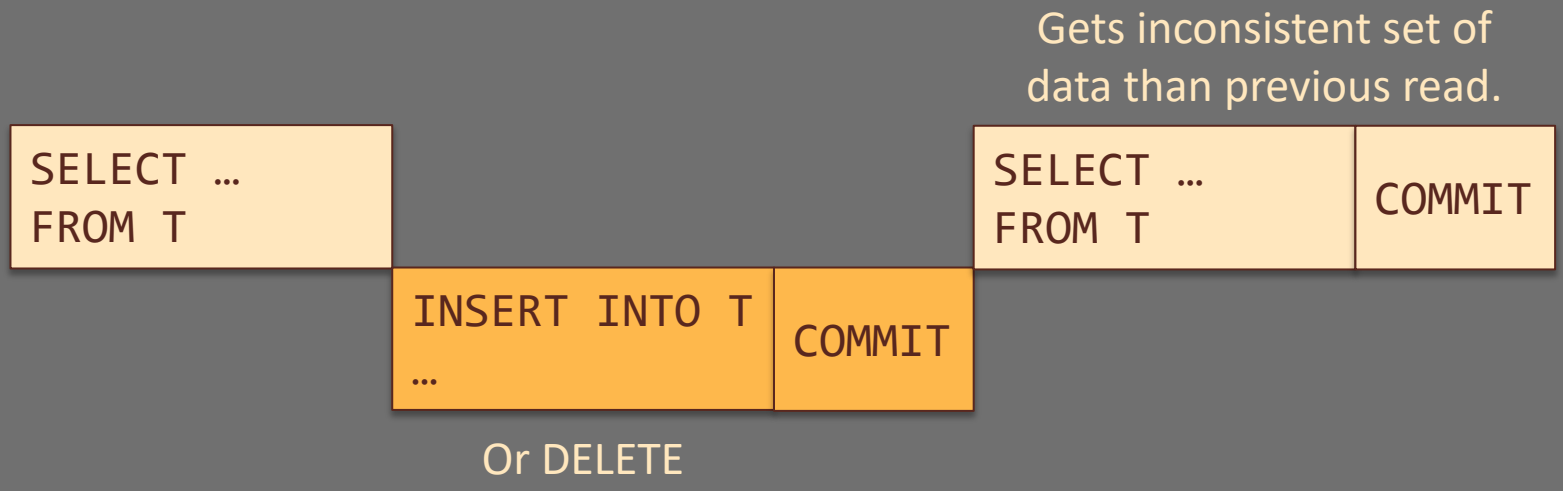


Gets different value than previous read.

Real-world scenarios:
Reload web page and …
- Price changes
- Item now sold out

Due to RW conflict.

Phantom read

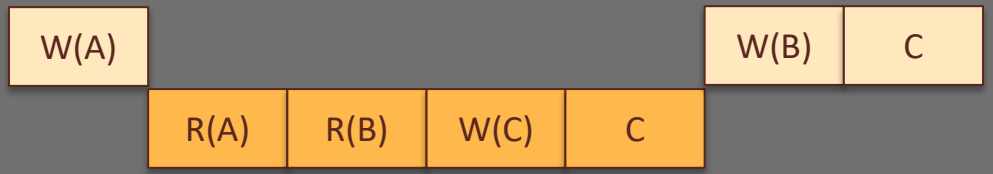


Real-world scenarios:

Reload web page and …

- Course enrollment changes.
- Available products changed.

Inconsistent read



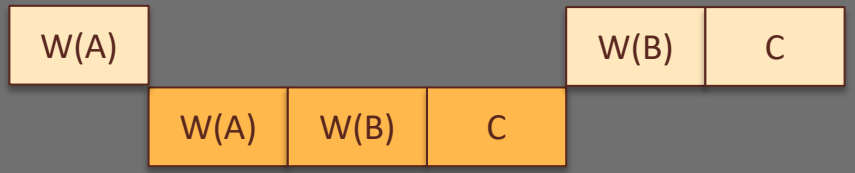
C depends on A,B.

C's value based upon incomplete commit.

Due to WR conflict.

Lost update

All blind writes



Have T_2's A and T_1's B.
Not serializable.

Due to WW conflict.

Conflicts and Anomalies

- Are they bad?

- Conflict serializable vs. anomaly serializable